

Nate Woodward

LC T P Spring symposium: Theoretical Physics and AI

Verifiable Tasks

Finetuning & Outcome Verification

1. Direct
calculations

2. Hidden-
coefficient
derivations

3. Ratio or
comparisons

4. Categorical /
logical
classifications



Code skeleton for model

Python Code Skeleton

```
def calculate_propagator_ratio(d: int) -> float:
    """
    Calculates the ratio  $R(d)$  of the self-contracted photon
    propagators in Feynman gauge to Landau gauge in  $d$ 
    spacetime dimensions.
    
$$R(d) = \frac{(D_{\{\mu\nu\}}^{\{F\}} D^{\{\mu\nu, (F)\}})}{(D_{\{\mu\nu\}}^{\{L\}} D^{\{\mu\nu, (L)\}})}$$

    Args:
        d (int): The number of spacetime dimensions.
    Returns:
        float: The numerical value of the ratio  $R(d)$ .
    """
    raise NotImplementedError
```


Reasoning finetuning is often driven by training with variable tasks



Yours are this in open-weight training by:



Qwen



Fine-Tuning & Outcome Verification

Reasoning fine-tuning is **often driven by training with outcome verifiable tasks**

You see this in open-weight trainings by:



Verifiable Tasks

- | | |
|-----------------------------------|--|
| 1. Direct calculations | 3. Ratio or comparisons |
| 2. Hidden-coefficient derivations | 4. Categorical / logical classifications |

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Aims of Our Datasets

✓ Wants:

DON'T Wants: